

Product Architecture & User Flow: StreamFlow

1. The Core Concept: "StreamFlow"

StreamFlow is an AI-powered, real-time invoice financing platform for Saudi micro-SMEs. It uses the **StreamPay API** to instantly pay suppliers on behalf of the SME, and then collects repayments via a "streaming" subscription model tied to the SME's daily POS sales.

2. User Registration Flow (The "Wow" Factor)

The goal is a 60-second onboarding experience, leveraging AI to eliminate manual data entry.

1. **Sign Up:** The SME owner (e.g., a cafe owner) enters their phone number and verifies via OTP.
2. **Document Upload (AI Magic):**
 - The user uploads a photo/PDF of their **Commercial Registration (CR)** and a recent **Supplier Quotation/Invoice** (e.g., for 50kg of coffee beans).
 - **AI Integration:** We use an LLM (like GPT-4o or Claude via Replit) to instantly extract:
 - CR Number, Business Name, Owner Name.
 - Supplier Name, Invoice Amount, Due Date, IBAN.
3. **Financial Connection:**
 - The user connects their POS system (e.g., Foodics) or Bank Account via Open Banking APIs (simulated for the hackathon).
 - **AI Risk Scoring:** A simple algorithm analyzes the last 3 months of POS data to determine a "Daily Repayment Capacity."
4. **Instant Approval:** The user receives an instant financing offer (e.g., "Approved for 15,000 SAR. Repayment: 500 SAR/day for 30 days").

3. The StreamPay API Integration (Perfecting the Rubric)

To score maximum points (15/15) on the "Stream API Integration" rubric, we must make it the *core* of the app.

Use Case 1: Paying the Vendor (The "Outflow")

- When the SME accepts the financing offer, StreamFlow uses the **StreamPay Payment Links API** or **Invoice API** to generate a payment directly to the supplier's IBAN.
- *Hackathon Demo*: We show the supplier receiving an instant notification that their invoice has been paid by StreamFlow on behalf of the cafe.

Use Case 2: Streaming Repayments (The "Inflow")

- This is where the "Stream" name shines. Instead of a bulky end-of-month loan payment, we set up a **StreamPay Subscription** for the SME.
- The subscription is configured to pull a small, manageable amount (e.g., 100 SAR) *daily* from the SME's account, mimicking a continuous "stream" of repayment tied to their daily coffee sales.
- *Hackathon Demo*: A live dashboard showing the repayment "streaming" in real-time, reducing the outstanding balance.

4. Technical Stack (Technical Execution - 30 Points)

- **Frontend**: React/Next.js (built via Replit) for a clean, modern UI (Tailwind CSS).
- **Backend**: Node.js/Express or Python FastAPI.
- **Database**: PostgreSQL (via Replit DB or Supabase) to store user profiles, invoices, and transaction history.
- **AI/ML**: OpenAI API for Document OCR (CR and Invoices) and basic risk scoring.
- **Payments**: **StreamPay API** (Core engine for all money movement).

5. UX/UI Design Principles (Product & UX/UI - 15 Points)

- **Mobile-First**: Cafe owners manage their business on their phones. The app must look like a native iOS/Android app.
- **"One-Click" Feel**: Minimize forms. If AI can read it from a document, don't ask the user to type it.
- **Visualizing the Stream**: The dashboard should have a beautiful, animated progress bar or "flowing water" graphic showing the daily repayments streaming back to the platform.

6. The "Live Demo" Strategy

1. **Start as the Cafe Owner**: Show the empty bank account but a busy cafe. Upload a CR and a 10,000 SAR invoice for a new espresso machine.

2. **Show the AI:** Watch the AI instantly parse the Arabic/English invoice and CR.
3. **The Approval:** Show the instant risk score based on simulated Foodics data.
4. **The StreamPay Magic:** Click "Finance & Pay Vendor." Switch screens to the Vendor's view to show the instant payment receipt via StreamPay.
5. **The Repayment Stream:** Fast-forward time (using a dev endpoint) to show the daily 333 SAR micro-payments streaming back via StreamPay Subscriptions.